



OPEN

Preparation of a novel 3D printed grinding wheel with linear cooling channels and study on the grinding performance

Jingjing Wu^{1,2}, Xuguang Zhao³, Qian Zhang^{4,5}✉, Shaohe Zhang^{4,5}✉, Xiangwang Kong^{4,5}, Linglong Rong^{4,5}, Yulu Li^{4,5}, Wenrui Gao^{4,5} & Zhuojun Xu¹

In this study, a novel type of diamond grinding wheel with linear cooling channels (GWLCC) is proposed, and an innovative manufacturing process is employed to develop this new grinding wheel. Grinding experiments conducted on four types of hard and brittle materials demonstrate that the grinding performance of GWLCC is better than that of conventional dense grinding wheels. Furthermore, the grinding performance of GWLCC was further investigated by comparing the abrasive ratio and working current under various grinding process parameters. The results indicate that the grinding efficiency of the grinding wheel is positively correlated with the rotating speed, achieving maximum efficiency at a rotating speed of 300 r/min. Additionally, variations in feed speed significantly affect the grinding force, but to protect the grinding equipment, the value shouldn't be too large.

Keywords Linear cooling channel, Diamond grinding wheel, Dual-filament 3D printing, Grinding performance, Grinding parameters

Grinding is a typical precision machining technology, which is often used as the final process of parts processing to ensure high dimensional accuracy and surface quality¹. Diamond grinding wheels are considered to be ideal tools for grinding glass, ceramics, gems, stone, and other hard and brittle materials due to their high hardness, strong wear resistance, and other excellent properties². They are widely used in aerospace, mold manufacturing, semiconductor manufacturing, optical glass manufacturing, and other fields³. Therefore, the research and development of high-performance grinding wheels plays a very important role in the advanced manufacturing industry. Metal-bonded diamond grinding wheels are most widely used in the field of grinding and precision machining of high-performance hard and brittle materials because of their advantages such as high bonding strength, long working life, good impact resistance, and wear resistance⁴. However, the lack of heat dissipation channels has caused a series of problems, limiting the application of diamond grinding wheels^{5,6}.

At present, the importance of pores or microgrooves has received increasing attention because they can provide larger chip removal space and coolant flow, effectively reduce the grinding temperature, and improve the chip removal ability of the grinding wheel⁷, thereby improving the machining quality of the workpiece. Therefore, structural designs such as grooves^{8,9}, holes^{10,11} and convex hulls^{12,13} are important means to improve the grinding performance of diamond grinding wheels. In this paper, a metal-bonded diamond grinding wheel with a linear cooling channel structure (GWLCC) in the entire working layer is proposed, to cut off the heat transfer route, improve the chip removal capacity of the grinding wheel, and improve the temperature field at the interface between the diamond layer and the workpiece. Our previous studies¹⁴ have shown that, compared with the dense diamond grinding wheels, under the same grinding conditions, GWLCC exhibits lower temperatures on the surface and subsurface of four workpiece materials (red sandstone, concrete, alumina ceramics, and optical glass), indicating that GWLCC can effectively improve the temperature field conditions in the grinding zone. However, the traditional hot-pressing sintering process limits the generation of micro-cooling channel

¹College of Civil Engineering, Hunan University of Technology, Zhuzhou 412007, Hunan, China. ²Technology Innovation Center for Deep Gold Resources Exploration and Mining, Ministry of Natural Resources, Weihai 264209, Shandong, China. ³Guangzhou Traffic Design and Research Institute Co., Ltd, Guangzhou 511430, China. ⁴Key Laboratory of Metallogenic Prediction of Nonferrous Metals and Geological Environment Monitoring, Ministry of Education, School of Geosciences and Info-Physics, Central South University, Changsha 410083, Hunan, China. ⁵School of Geosciences and Info-Physics, Central South University, Changsha 410083, Hunan, China. ✉email: 1041117656@qq.com; zhangsh@csu.com

structures for grinding wheels, especially metal-bonded grinding wheels. The development of dense diamond grinding wheels is restricted by the problems of large grinding force, high grinding temperature, and grinding wheel blockage¹⁵.

As the main manufacturing process for grinding wheels, the hot pressing method has great limitations¹⁶. On the one hand, the high temperature and pressure environment hinder the formation of micro-channels in the grinding wheel. On the other hand, the shape of the flow channel cannot be accurately controlled and designed. Therefore, the controllable design and accurate preparation of micro-channel structure are of great significance for metal-bonded grinding wheels, but it is difficult to achieve with the traditional hot pressing method.

At present, mechanical dressing and laser dressing are mainly used for structuring the surface or substrate of grinding wheels. As early as 1977, Nakayama et al.¹⁷ prepared the grooved structured grinding wheel by the mechanical method, and since then, the mechanical methods have continued to develop as the main method for preparing grooved structured grinding wheels. Azarhoushang et al.¹⁸, Denkena et al.¹⁹, and Zhang et al.²⁰ respectively used a grinding wheel dresser, fly cutting method, and abrasive water jet method to prepare grooved structured grinding wheels. However, the geometric shape accuracy of the surface structure is highly dependent on the preparation tool when the mechanical method is used to prepare the grooved structured grinding wheel, which limits the complexity of the prepared geometric shape. Therefore, the mechanical method is generally suitable for the preparation of the macro-structured grinding wheel. As the structure of grinding wheels develops towards microscale, multi-dimensional, and multi-functional directions, the laser method is gradually becoming the main preparation method for microstructured grinding wheels. Khangar²¹, Li²², and Fu et al.²³ successively applied laser processing technology to the preparation of structured grinding wheels, and respectively prepared grooved alumina grinding wheels, resin-bonded diamond grinding wheels, etc. The laser method is very suitable for the processing of structured grinding wheels because of its high flexibility, non-contact, and other characteristics. However, the laser can easily cause thermal damage to the diamond particles²⁴, reducing the mechanical properties of diamond abrasive particles. Moreover, due to the lack of suitable laser processing parameters and laser path planning, it is difficult to process microstructures with controllable size and high precision²⁵. In addition, the depth of the microstructure prepared by the existing process is usually only hundreds of microns, and the microstructure is inevitably worn with the progress of grinding, so the cooling and chip removal function of the microstructure is ineffective. Therefore, there is an urgent need for a new technology that can reduce thermal damage and improve structural integrity and depth to promote the comprehensive improvement of diamond grinding wheel performance.

Our team has been working on the additive manufacturing of metal-bonded diamond tools for a long time. In 2018, the application of additive manufacturing technology in diamond tool manufacturing was discussed²⁶. In 2019, a new 3D-printed diamond-impregnated bit with a grid-shaped matrix was prepared by SLS technology²⁷, and its rock-breaking mechanism was further studied in 2020²⁸. In 2021, in view of the problem of diamond thermal damage that is prone to occur in the SLS process, a new idea for preparing diamond tools was put forward, that is, fused deposition modeling and sintering (FDMS) process²⁹. High-performance diamond composite filaments suitable for this technology have been developed^{30–32}, and high-precision, high-performance diamond ultra-thin slices of different specifications have been successfully prepared using this technology. These studies have shown that FDMS technology has been successfully applied in the field of diamond tool manufacturing. The FDMS process transfers the thermal processing step to the sintering step, which can achieve the purpose of easy control of thermal stress. Therefore, the application of FDMS technology in the manufacturing of diamond grinding wheels with micro-channels can realize the controllable design and accurate manufacturing of micro-structured metal-bonded diamond grinding wheels, thus getting rid of the limitations of traditional manufacturing methods.

In addition, the reasonable determination of grinding process parameters plays an important role in ensuring processing quality, improving productivity and reducing production costs³³. Chen³⁴ used a single-layer diamond grinding wheel prepared by vacuum brazing to grind alumina materials at three different grinding speeds. With the increase of the rotation speed of the outer edge grinding wheel, the grinding force decreased, but the specific energy of wear and the grinding temperature increased. Liu et al.³⁵ found that the grinding parameters of diamond grinding wheel had similar effects on the surface roughness and subsurface damage depth of silicon nitride after grinding. With the increase of grain size and grinding wheel speed, the surface roughness and subsurface damage depth decreased, while with the increase of workpiece speed and grinding depth, the surface roughness and subsurface damage depth increased. Agarwal et al.³⁶ used diamond grinding wheels to grind silicon carbide materials under different grinding parameters. The results showed that cutting depth, feed rate, sand size, sand density, and other parameters were the main factors affecting the surface integrity. Qiao et al.³⁷ studied the effects of machining parameters on surface roughness in ultrasonic vibration grinding of hot-pressed silicon nitride and found that spindle speed has the greatest influence on surface roughness, followed by feed speed, cutting depth, and amplitude. The above conclusions fully confirm that the grinding process parameters can have an important influence on the material removal rate and product processing quality. In particular, the grinding wheel speed and feed speed directly control the material removal mechanism and have the most significant impact on the grinding work.

To sum up, a metal-bonded diamond grinding wheel with the linear cooling channels is proposed in this paper, which is realized by using the dual-filament FDMS 3D printing technology. In addition, the grinding experiments on four kinds of hard and brittle materials including red sandstone, concrete, alumina ceramic plate, and optical glass, verify the improvement effect of the linear flow channel structure on the grinding performance of the new diamond grinding wheel. At the same time, based on the above-optimized preparation process parameters, the influence of rotation speed and feed speed on the grinding performance of the grinding wheel is further discussed.

Preparation of GWLCC
Structure of the GWLCC

The newly developed diamond grinding wheel with linear cooling channels includes a working layer and a metal substrate (Fig. 1). The entire working layer is uniformly distributed with multiple linear cooling channels, which can not only improve chip removal efficiency but also interrupt the heat pathway during operation. This design significantly reduces thermal damage to the diamond, thereby extending the service life of the grinding wheel and improving the surface quality of the workpiece. To facilitate processing, this new grinding wheel has an inner diameter of 52 mm, an outer diameter of 64 mm, a total height of 4 mm, and flow channel dimentions: width, depth, and inclination of 1.5 mm, 2 mm, and 30°, respectively.

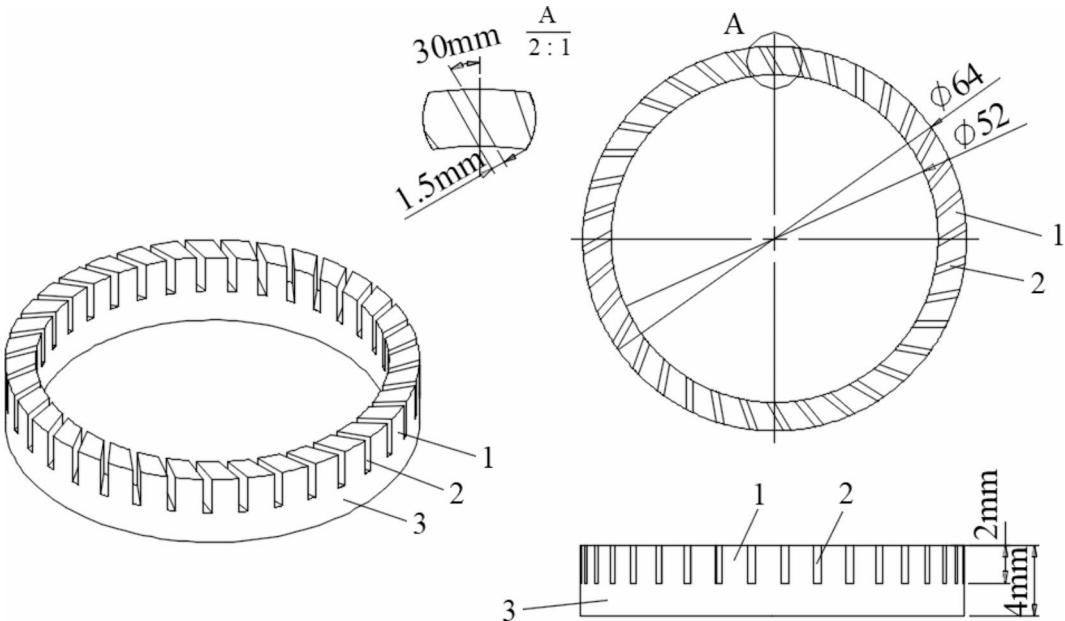


Fig. 1. Structure diagram of GWLCC. 1-working layer, 2-linear cooling channel, 3-metal substrate.

Preparation of the diamond grinding wheel with linear cooling channel

The preparation of GWLCC was carried out according to the formula in Table 1. According to previous studies, the preparation of the grinding wheel needs to be completed through a series of experimental processes. The experimental procedures should be strictly followed in the preparation process, otherwise the performance of the grinding wheel will be adversely affected. The dual-filament FDMS printing process involves two nozzles in the printing of parts, realizing the simultaneous printing of two materials. In this study, it is used to simultaneously print the working layer and the linear cooling channel. The preparation process of the GWLCC is shown in Fig. 2, and the preparation process parameters are shown in Table 2.

The preparation process is divided into burdening, internal mixing, granulation, filament drawing, dual-filament 3D printing, degreasing, and high-temperature sintering. Firstly, according to the designed formula, the raw materials for forming the working layer of the grinding wheel and the channel were weighed respectively by using an electronic scale with an accuracy of 0. 01 g. Then the two parts of materials were placed in an internal mixer (KH type internal mixer) twice, and the materials were uniformly stirred and bonded into a lump body. After internal mixing, the bonded block material is broken into fine particles with uniform volume by a

	Volume ratio	Raw materials	Volume ratio	Particle size (mesh number)
Grinding wheel body material	50%	Diamond powder (MBD ₆)	30%	100/120#
		CuSn15 alloy powder	70%	300#
	50%	Binder (high-temperature resistant resin, thermoplastic elastomer, and polyolefin)	/	/
Channel forming material	45%	Graphite powder	45%	400#
		Alumina powder	55%	600#
	55%	Binder(high-temperature resistant resin, thermoplastic elastomer, polyolefin and surfactant)	/	/

Table 1. Composition and proportion of GWLCC for 3D printing.

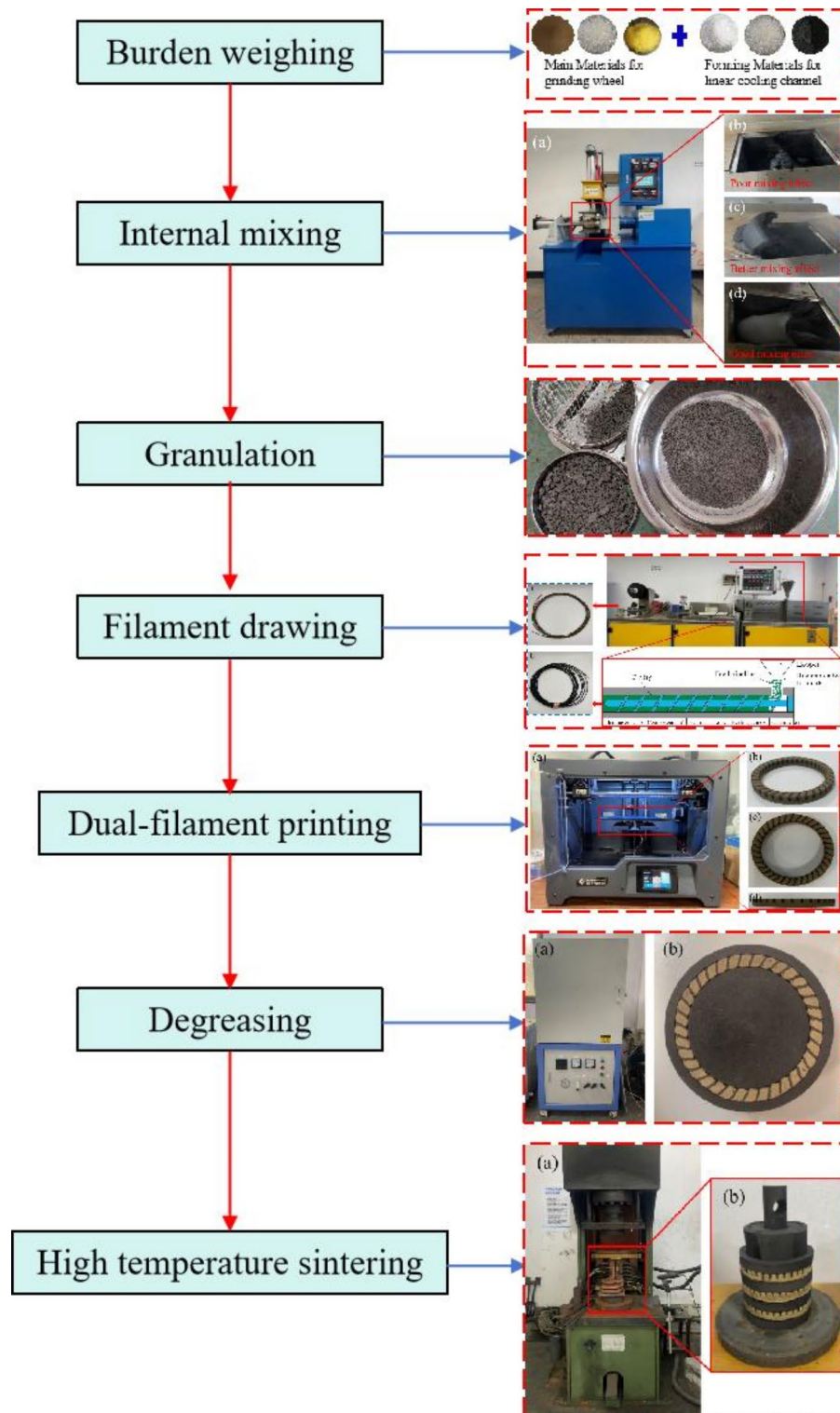
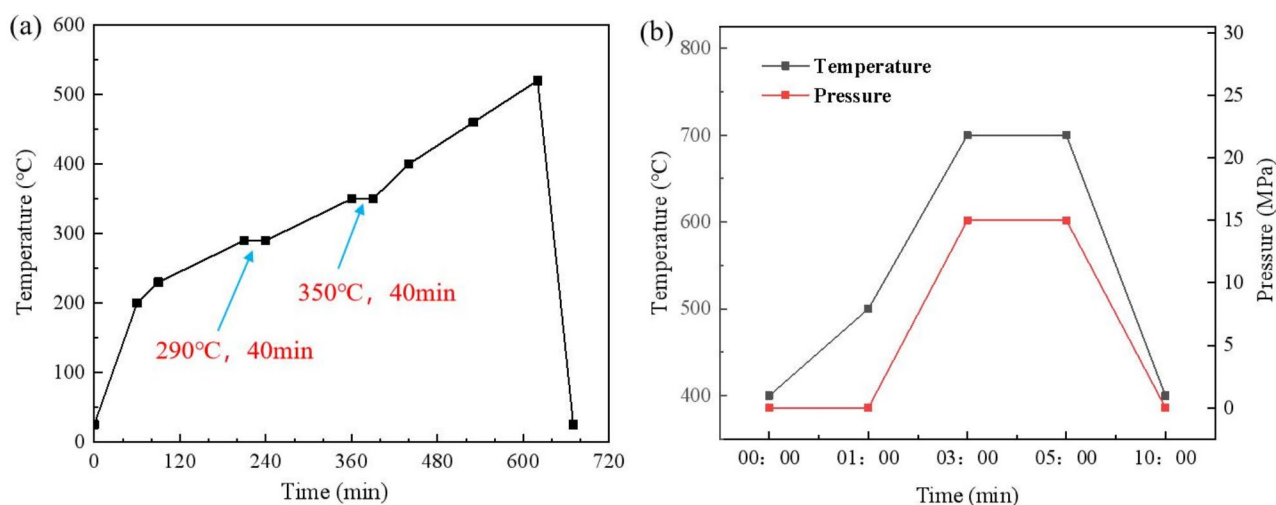


Fig. 2. Process flow of GWLCC manufactured by dual-filament FDMS technology.

granulator, and then filled into the hopper of a single-screw extruder (SJ25 series single-screw extruder), and finally rolled into filaments under the action of a traction wheel to prepare the filaments for forming the working layer and the flow channel respectively. Then, start the dual-nozzle 3D printer (Wolverine 2pro double-nozzle 3D printer), import the grinding wheel model that the printer can recognize, and set the printing parameters. The two types of filaments were fed into the filament feeding pipe of the dual-nozzle 3D printer respectively. After the filament feeding was completed, the nozzle was printed layer by layer according to the working path of the design model, and the diamond grinding wheel green body with linear cooling channels was prepared.

Internal mixing	Temperature	Rotor speed		Heating cavity temperature
	190 °C	25 r/min	Filament drawing	
Printing	Nozzle temperature	Platform temperature	Print speed	Fill Mode/Density
	210	95	30 mm/s	Linearity/100%

Table 2. Process parameters of GWLCC prepared by dual-filament FDMS technology.**Fig. 3.** Curve of (a) degreasing process and (b) sintering process.

The green body was embedded into a degreasing mold for limiting deformation, and then they were placed into a degreasing furnace (a high-temperature vacuum atmosphere furnace) together, and the degreasing process parameters were set to start degreasing. After being degreased, the green body was put into a sintering mold, and the sintering parameters were set. Then the sample was placed in a medium-frequency sintering furnace to complete sintering. Finally, the finished diamond grinding wheel containing linear cooling channels is obtained after trimming.

During the degreasing process, the binder in the green body gradually changes from the initial solid phase to the liquid phase and then to the gas phase. Therefore, the microstructure, density, and porosity of the grinding wheel vary greatly due to the rapid heating rate, which also makes the performance of the grinding wheel different. The heating rate will affect the quality of the grinding wheel. In the initial stage of the experiment, the degreasing process was set at a fixed heating rate of 1 °C/min from room temperature to 520 °C. The surface of the resulting green body was bulged and greatly deformed, and the degreasing effect was poor. According to the existing test conditions and experience, degreasing is performed in different heating processes, the degreasing rate is calculated and the quality of the green body is compared, and the optimal degreasing process is determined, as shown in Fig. 3(a). A two-stage heat insulation process is set during the degreasing process, wherein the temperature of the first insulation stage is 290 °C to soften the binder, and the temperature of the second insulation stage is 350 °C to prevent defects in the grinding wheel due to insufficient phase change or rapid phase change. In addition, in the process of degreasing, a matching graphite mold (including an outer mold and a core mold) is required for degreasing the diamond grinding wheel to limit the shrinkage of the inner diameter and the expansion of the outer diameter of the diamond grinding wheel.

During the sintering process, if the pressure is too high, it is easy to cause a serious overflow phenomenon and may also cause physical damage to the grinding wheel, while if the pressure is too low, the grinding wheel can not be sintered densely. In addition, excessively high temperature can result in significant loss of low-melting-point metals and may also damage the diamond particles. Conversely, if the temperature is too low, the metal material will not melt adequately, preventing effective metallurgical bonding. Therefore, both excessively high and low temperature can compromise the performance of the grinding wheel. After consulting relevant literature and experimental verification, the sintering process is shown in Fig. 3(b).

By optimizing the process route, the finished GWLCC is finally obtained after trimming, as shown in Fig. 4. It can be seen from Fig. 4 that the surface of the grinding wheel is flat and the chip removal channel is well formed. It can be seen from the figure that the surface of the grinding wheel is flat and the chip removal water tank is well-formed. In addition, the microscopic morphology of the surface of the grinding wheel after grinding reveals that the diamonds in the working layer are uniformly distributed and well exposed. Observations of individual diamond particles indicate that the matrix exhibits a strong adhesion to these particles, with no visible cracks present between the diamond and the matrix. The performance parameters of the grinding wheel are presented

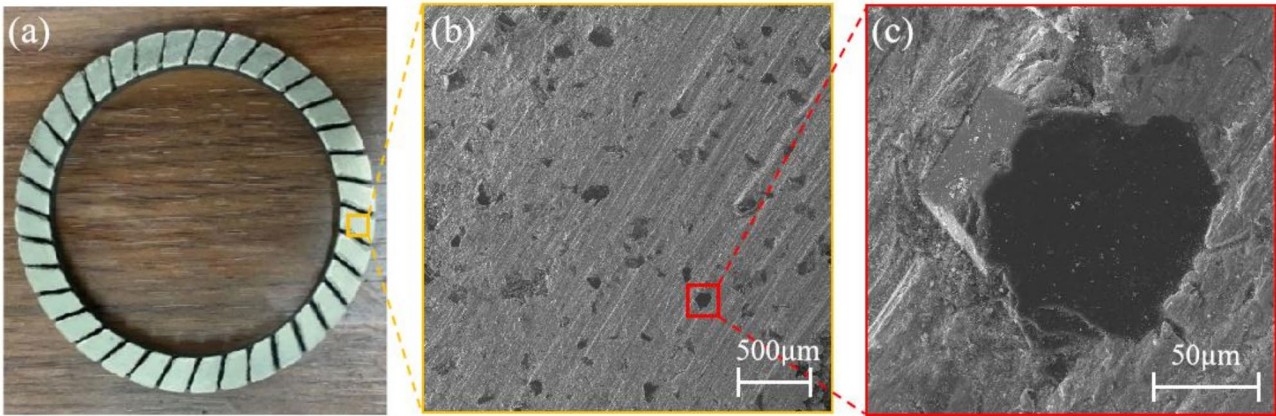


Fig. 4. 3D printing diamond grinding wheel. (a) 3D printed GWLCC, (b) matrix surface morphology of yellow region, and (c) microscopic morphology of single diamond particle.

Relative density	Hardness/HRB	Bending strength/ MPa	Holding force coefficient
92.67	79.5	509.12	61.58%

Table 3. Performance parameters of GWLCC.

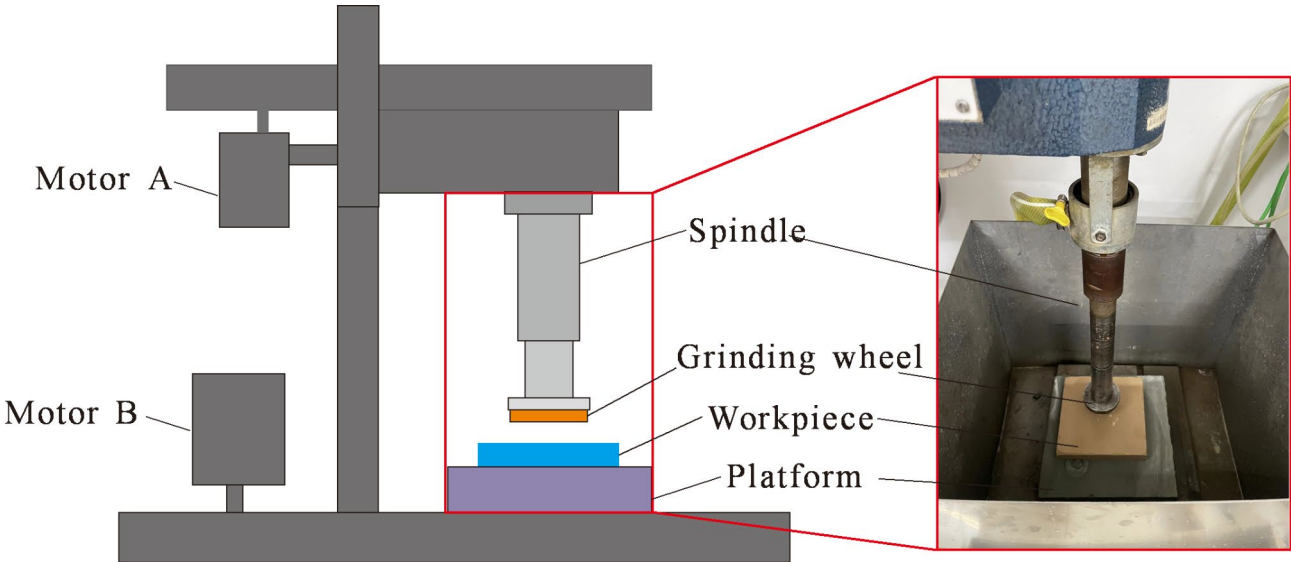


Fig. 5. Schematic diagram of grinding equipment.

in Table 3. These findings demonstrate that the diamond grinding wheel preparation scheme proposed in this paper is feasible.

Laboratory validation of the linear cooling channel structures
Testing method

In order to verify whether the linear cooling channel structure can improve the grinding performance of the diamond grinding wheel, we printed GWLCC and conventional dense diamond grinding wheel without channels according to the optimized formula and process parameters in Tables 1 and 2, and carried out grinding experiments to evaluate the feasibility of the linear cooling channel structure.

The grinding test platform is shown in Fig. 5. The processing objects of this test are red sandstone, concrete, alumina ceramic plate, and optical glass, and the specific performance parameters of the workpieces are shown in Table 4. In the grinding experiment, the diamond grinding wheel was fixed on the grinding equipment using a metal-specific super glue, and the 100 mm × 100 mm plane of the four workpieces was ground for 20 s. The grinding process parameters are shown in Table 5. The evaluation indexes are the grinding efficiency and grinding

Material	Elastic modulus/ GPa	Poisson's ratio	Mohs hardness	Density/(g/cm ³)	Specific heat capacity/J/ (kg·K)	Coefficient of thermal expansion/ m/m·k	Conductivity/ W/m·K
Red Sandstone	11	0.26	6	2.23	630	7.8*10 ⁻⁶	1.72
Concrete	28	0.20	6.5	2.01	970	10*10 ⁻⁶	1.30
Alumina ceramic	300	0.20	8.5	3.70	750	7.2*10 ⁻⁶	22.00
Quartz glass	76	0.17	7	2.20	964	0.55*10 ⁻⁶	1.55

Table 4. Performance parameters of four workpieces. Tips: data sources are produced by Jiaxiang Jinxu Stone Industry Co., Ltd(red sandstone), Cangzhou Fuyi building materials and instruments (Concrete), Guangzhou Huameng new material factory (Alumina ceramic) and Donghai County Dongrong quartz products Co., LTD (Quartz glass).

Parameter type	Configuration parameters
Grinding method	Flat grinding
Rotating speed	270r/min
Feed speed	Grade I
Type of coolant	Water-based coolant
Grinding time	20s

Table 5. Grinding test parameters.

force of two diamond grinding wheels grinding four workpieces. The surface morphology of the grinding wheels is observed with the help of scanning electron microscopy to further evaluate the grinding effect.

(1) Abrasion ratio

Abrasion ratio G in this study³⁸ is defined as the ratio of the mass of workpiece material removed to the wear mass of the grinding wheel within a single grinding time of 20 s. The mass change of the workpiece and the grinding wheel before and after grinding is measured with a precision electronic scale (JA10003B with an accuracy of 1 mg). The measurements were repeated 5 times, and the average value was taken as the test result. The calculation formula of the abrasion ratio G is as follows:

$$G = \frac{M_w}{m_s} \tag{1}$$

$$M_w = M_1 - M_2 \tag{2}$$

$$m_s = m_1 - m_2 \tag{3}$$

where M_w is the mass of the workpiece material removed in a certain time (g), m_s is the mass of the grinding wheel wear in a certain time (g), M_1 is the mass of the workpiece before grinding (g), M_2 is the mass of the workpiece after grinding (g), m_1 is the mass of the grinding wheel before grinding (g), m_2 is the mass of the grinding wheel after grinding (g).

(2) Grinding force

In this paper, the indirect analysis method is used to study the grinding force variation in the grinding process. Based on the existing research, the working current of the spindle is selected as the reference index of the grinding force variation. The working current is monitored in real time by connecting the digital ammeter in series to the working circuit of the spindle motor. And the average working current was taken as the test result.

Test results and analysis

Effect of linear cooling channel on grinding efficiency of diamond grinding wheel

Figure 6 illustrates the abrasive ratio of two grinding wheels utilized for processing different workpieces. It is evident that, in comparison to the diamond wheel without cooling channel, the GWLCC achieves a significantly higher abrasive ratio when grinding red sandstone, concrete, alumina ceramics, and optical glass. The abrasive ratio increases by 32–232.67%, indicating that the optimized structure enhances the grinding efficiency of the diamond wheel. In addition, there are notable differences in the abrasive ratios of the two grinding wheels when processing different materials. During the concrete grinding tests, the disparity in abrasive ratio between the two grinding wheels was most pronounced, reaching approximately 35. Conversely, when grinding alumina ceramics, the abrasive ratios of both grinding wheels are relatively similar. However, a certain degree of difference still persisted. The smaller discrepancy may be attributed to the water-based coolant during low feed speed grinding conditions, which resulted in slippage of both diamond grinding wheel structures.

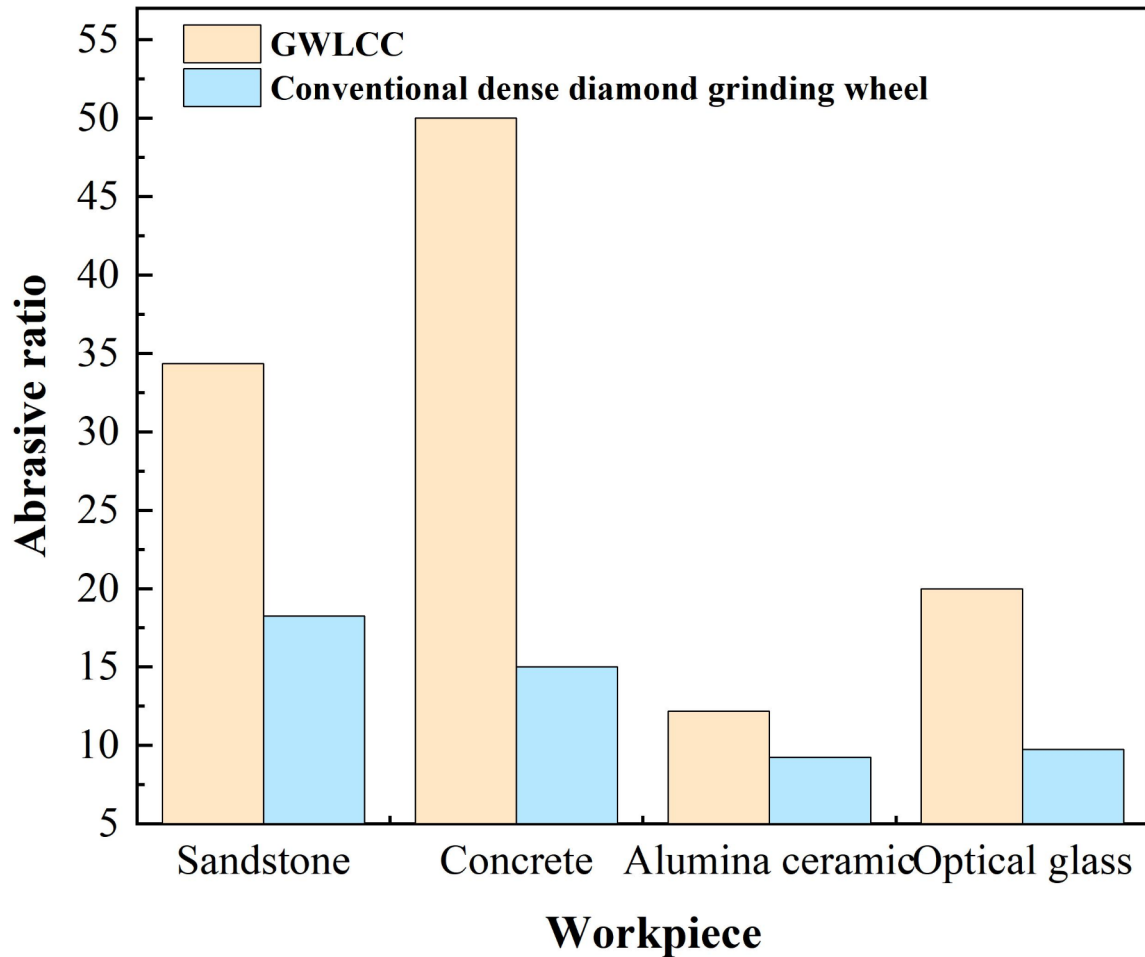


Fig. 6. Abrasion ratio of diamond grinding wheel with different structure.

Figure 7 shows the microscopic morphologies of the surfaces of the two diamond grinding wheels. It can be seen from Fig. 7 that the surface of the grinding wheel without the cooling channel exhibits numerous grinding chips of varying sizes adhered to it, leading to increased susceptibility damaged for the diamond particles. In contrast, the surface of the GWLCC shows minimal adherence of grinding chips. This observation indicates that the linear cooling channel structure facilitates effective removal pathways for grinding chips, thereby enhancing the chip removal capacity of the grinding wheel and reducing contact between diamond particles and debris. Consequently, this improvement contributes to an increase in the effective grinding capacity of the grinding wheel. Furthermore, when combined with the numerical simulation results pertaining to temperature fields¹⁴, it is evident that this structural design can significantly lower temperature within the grinding zone. This deduction is beneficial in mitigating thermal damage to diamonds and preserving their mechanical properties. Therefore, this optimized structure demonstrates a substantial potential for improving the grinding efficiency of the diamond grinding wheel.

Effect of linear cooling channel on grinding force of diamond grinding wheel

Figure 8 shows the current value and the variation in current for two diamond grinding wheels during the grinding process. It can be seen from Fig. 8 that compared to the grinding wheel without channel, the working current of GWLCC is reduced by 6.22–14.64%. This reduction indicates that the design of the linear channel structure effectively minimizes the grinding force required by the grinding wheel. That is because the working surface area of the diamond grinding wheel without flow channels is larger than that of the GWLCC, so the grinding force will also be greater. Furthermore, when dealing with more challenging workpiece materials, it can be anticipated that the disparity in grinding forces between these two structures may become even more pronounced. On the other hand, it can also be observed that when grinding alumina ceramics, the difference in working current between the two grinding wheels is the largest. This indicates that the grinding force difference is the largest at this time. According to the previous numerical simulation calculations¹⁴ and the abrasive ratio

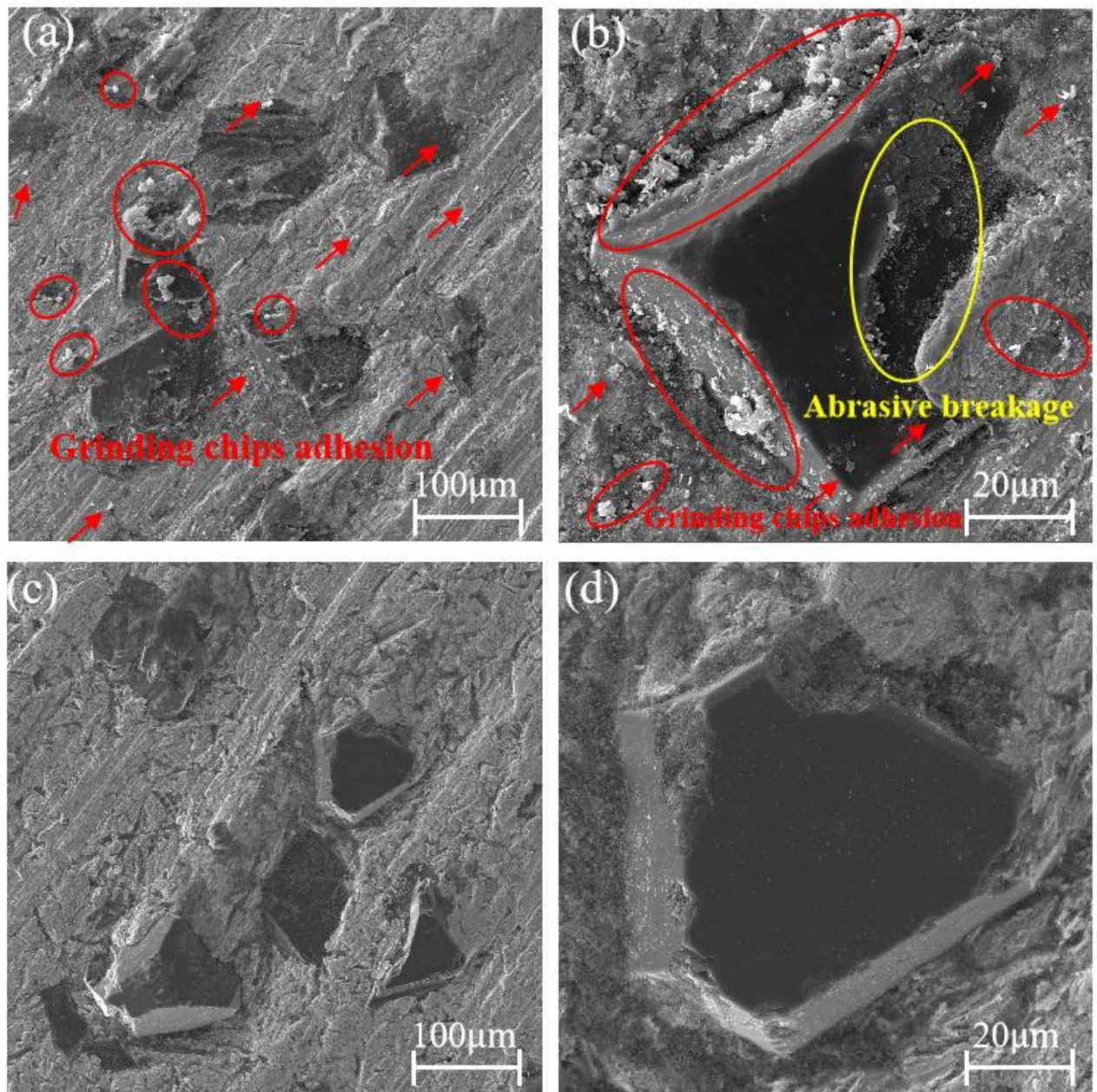


Fig. 7. (a) and (b) microtopography of diamond grinding wheel without flow channel, (c) and (d) microtopography of GWLCC.

results, it has been established that when machining hard and smooth workpiece materials such as alumina ceramics and optical glass, the diamond particles experience rapid wear and significant passivation. This phenomenon leads to a continuous increase in the contact area between the grinding wheel and the workpiece. Consequently, there is an escalation in grinding force, which is reflected by a continuous increase in current values.

Effect of grinding process parameters on grinding performance of GWLCC

Test materials and test method

The GWLCC was prepared according to the optimized formula and process technology outlined in the previous section. During the grinding test, the GWLCC was installed on the grinding test machine shown in Fig. 5 to grind a 100 mm × 100 mm plane of red sandstone, with a single grinding duration of 40 s. To evaluate the influence of rotation speed and feed speed on the grinding performance of diamond grinding wheel, an experimental approach utilizing a single variable method was employed. Five groups of rotation speeds were set, namely 180 r/min, 210 r/min, 240 r/min, 270 r/min and 300 r/min. Additionally, three levels of feed speed were defined as Grade I (small), Grade II (medium), and Grade III (large). The rotational direction and feed direction of the

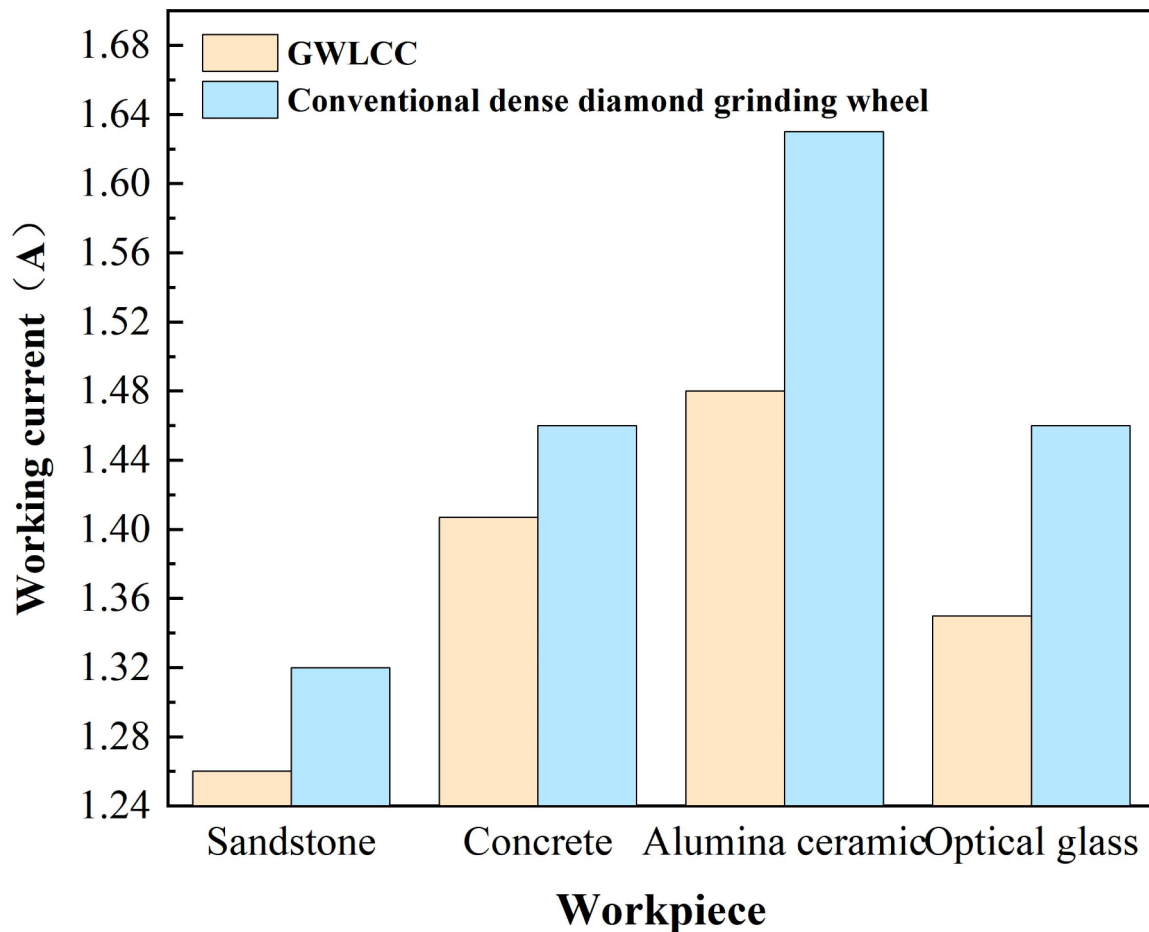


Fig. 8. Influence of diamond grinding wheel structure on working current.

spindle are illustrated in Fig. 9, while Table 6 presents the parameters for the grinding process. The evaluation indexes for both sets of tests include grinding efficiency and grinding force, and these calculations follow the methodology described in above section. This study also investigates variations in abrasive ratio and working current under different combinations of rotation speeds and feed speeds to explore general trends within these parameters.

Test results and analysis

Influence of mechanical rotation speed on grinding performance of grinding wheel

Table 7 presents the wear characteristics of diamond grinding wheel during the machining of red sandstone at different rotation speeds. The test data show that, for a constant working duration, an increase in rotational speed leads to a decrease in the wear mass of the diamond grinding wheel and an increase in material removal mass. Specifically, as the grinding speed rises from 180 r/min to 300 r/min, the removal mass of red sandstone increases from 11.34 g to 13.69 g. This phenomenon can be attributed to the inverse relationship between the time required for one complete rotation of the grinding wheel and its rotational speed. When maintaining a consistent grinding duration, increasing the speed of the grinding wheel enhances the number of effective abrasive grains engaged in grinding work per unit time, thereby resulting in an increased removal mass of red sandstone.

During the experiment, it was observed that the wear mass of the grinding wheel exhibited a decreasing trend with an increase in grinding times. This phenomenon may be attributed to the metal-bonded diamond grinding wheel, which utilizes CuSn15 alloy powder as its matrix. In this configuration, the diamond grains could be firmly fixed and seldom fell off during the grinding process. The primary mode of abrasive wear is tip wear. Initially, during continuous grinding, abrasives with sharp cutting edges experience a reduction in their exposure height due to the tip wear, as shown in Fig. 10. Furthermore, vibrations from the grinding equipment can also induce adaptive wear on both the diamond abrasives and their matrix. Once the grinding interface adjusts to these vibrations, the diamond grinding wheel transitions into its normal operational stage.

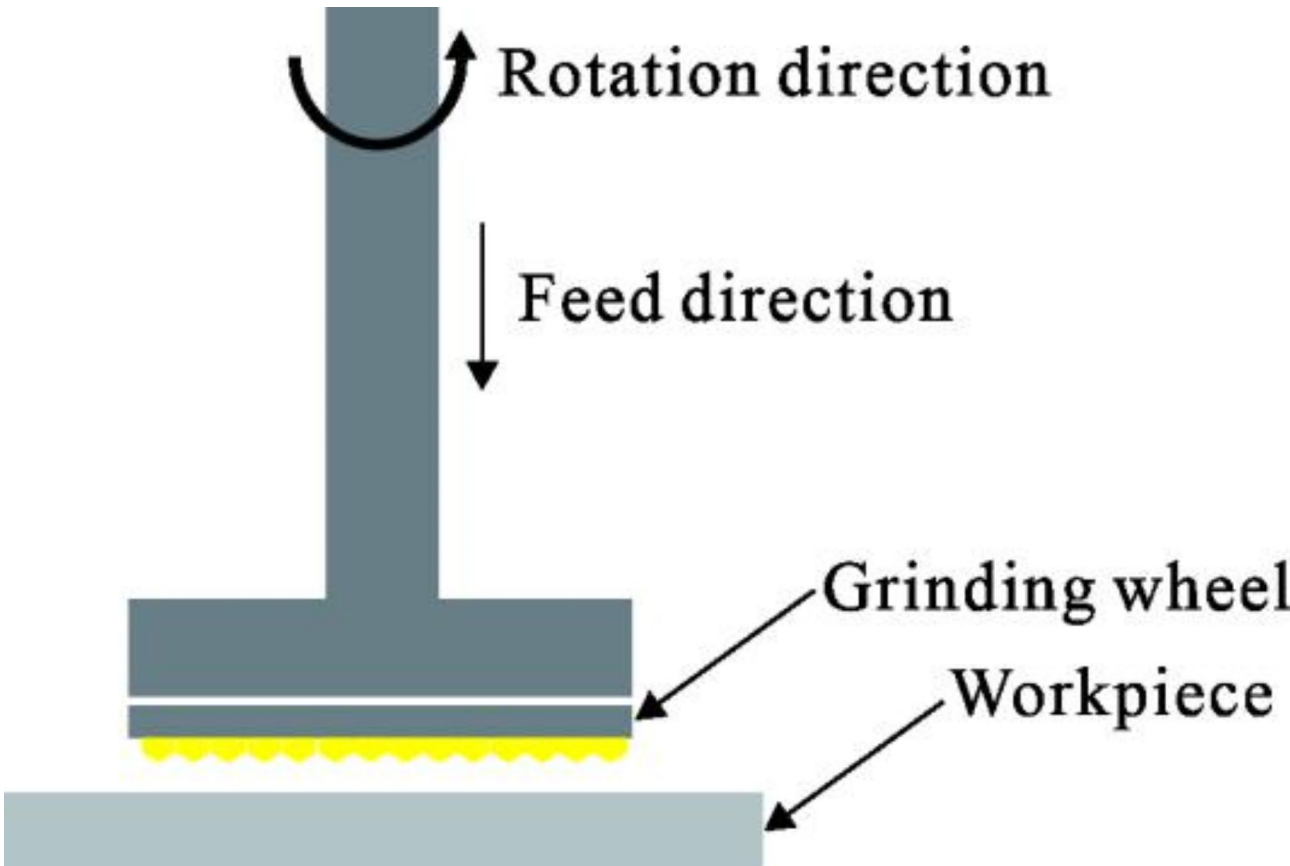


Fig. 9. Schematic diagram of spindle rotation and feed direction.

No.	Rotating speed/ (r/min)	Feed speed	Grinding time/s	Grinding method	Type of coolant
I	1 180	Grade I	40	Flat grinding	Water-based coolant
	2 210				
	3 240				
	4 270				
	5 300				
II	6	Grade I	40	Flat grinding	Water-based coolant
	7 270	Grade II			
	8	Grade III			

Table 6. Grinding test parameters.

Speed/ (r/min)	Feed speed/grade	Grinding wheel mass (m ₁)/g	Grinding wheel mass (m ₂)/g	Wear mass (m _s)/g	Sandstone quality (M ₁)/g	Sandstone quality (M ₂)/g	Mass removed (M _w)/g
180	I	437.00	436.62	0.38	1067.92	1056.58	11.34
210	I	436.62	436.50	0.12	1055.98	1044.37	11.61
240	I	436.50	436.40	0.10	1054.37	1042.40	11.97
270	I	436.40	436.33	0.07	1051.31	1038.54	12.77
300	I	436.33	436.27	0.06	1058.54	1044.85	13.69

Table 7. Abrasion of GWLCC machining red sandstone at different rotational speeds.

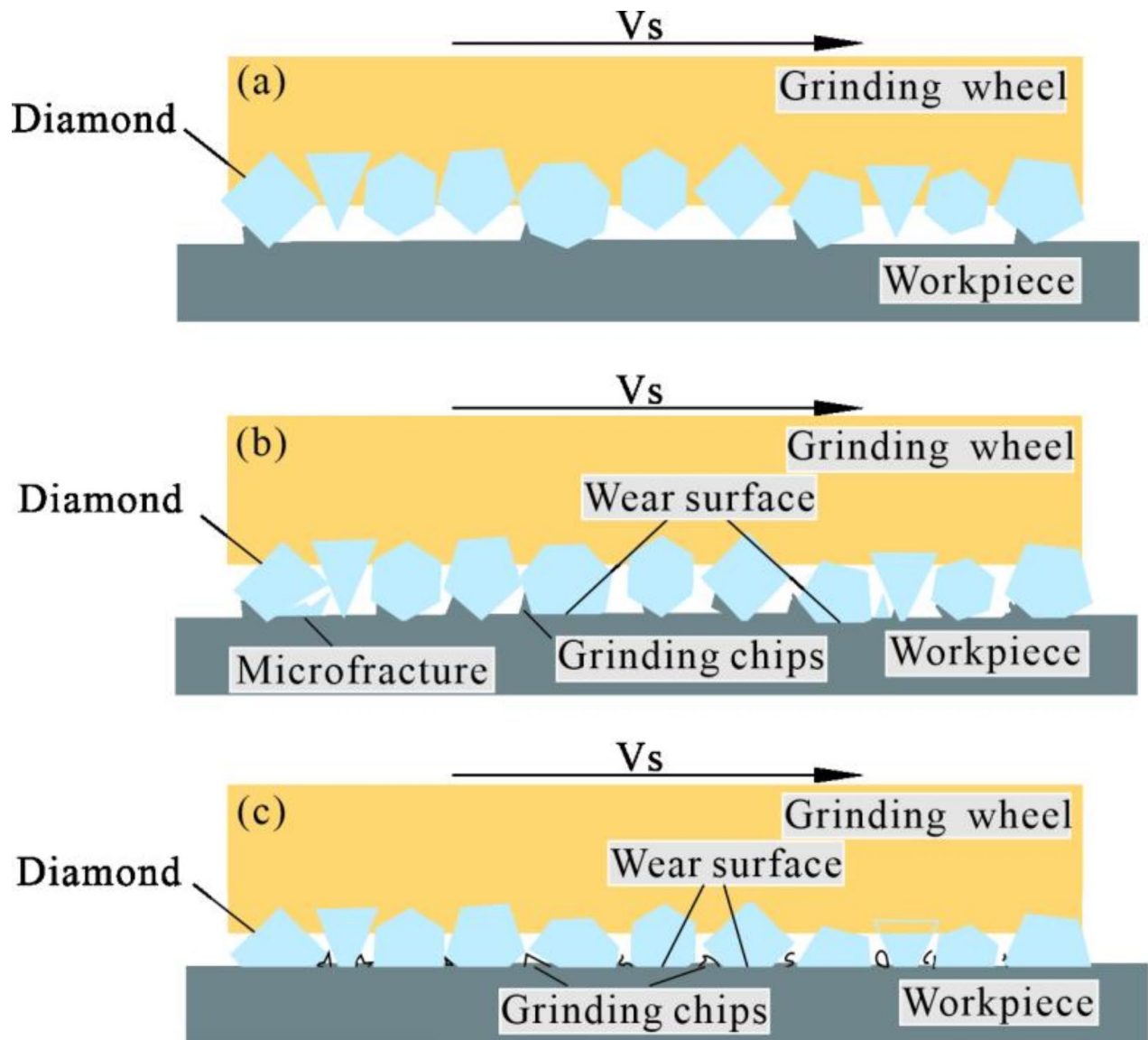


Fig. 10. Abrasive state of diamond particles³⁹ during (a) early stages of the grinding process, (b) intermediate stages of the grinding process and (c) late stages of the grinding process.

Figure 11 illustrates the relationship between the rotational speed of the grinding wheel and the abrasive ratio. As depicted in Fig. 11, it is evident that the abrasive ratio increases continuously with an increase in the diamond grinding wheel speed. This is because that the grinding work involves the mechanical removal of the workpiece material by diamond abrasive grains under conditions of high-speed rotation. When the rotating speed of the grinding wheel increases, there is a corresponding increase in the frequency at which diamond grains remove material from the workpiece within a given time frame. Consequently, this enhances both the working efficiency of the grains and their material removal capacity, leading to an increased abrasive ratio for the grinding wheel.

The working current data of diamond grinding wheel used in the grinding of red sandstone is shown in Table 8. It can be seen that the current increases with the increase of the spindle speed. On one hand, the working current of the motor is directly correlated with its power output. Given that a digital ammeter is connected in series with the motor controlling the rotation of the grinding wheel spindle, an increase in spindle speed results in a corresponding rise in motor power. At this point, since voltage remains constant, there is a proportional increase in working current. On the other hand, during the grinding process, as the grinding time increases, wear continues to increase and abrasive particles gradually become passivated. This phenomenon leads to an increase in grinding force and a reduction in grinding efficiency, resulting in a continues increase in working current. By analyzing the experimental data and conditions, it can be seen that the series connection of the digital ammeter is primarily responsible for this increase in current. Specifically, when speed increments are applied regularly at intervals of 300r/min, the increase in working current still exhibits a discernible pattern.

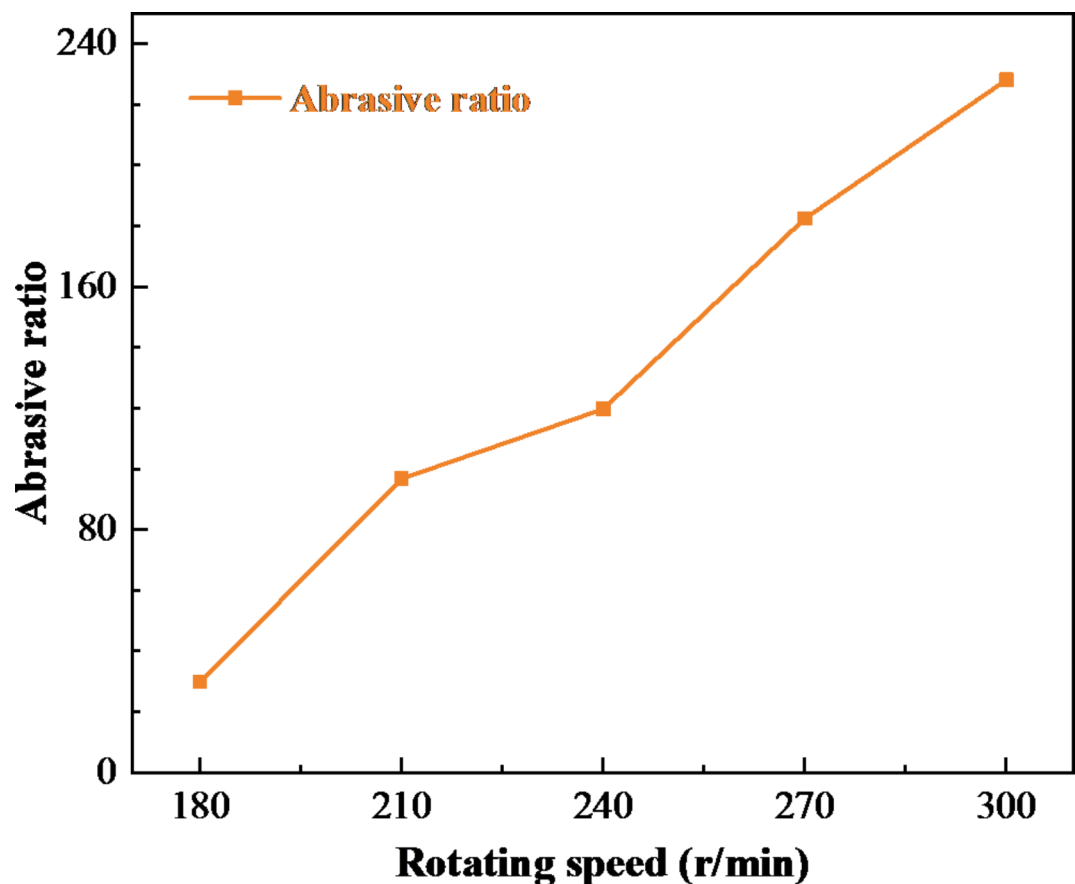


Fig. 11. Grinding wheel rotating speed vs. abrasive ratio.

Speed/(r/min)	Feed speed/grade	Working current/A
180	I	1.160
210	I	1.310
240	I	1.420
270	I	1.520
300	I	1.660

Table 8. Working current of diamond wheel grinding red sandstone at different rotational speeds.

Feed speed/grade	Rotating speed/(r/min)	Grinding wheel mass (m_1)/g	Grinding wheel mass (m_2)/g	Wear mass (m_s)/g	Sandstone quality (M_1)/g	Sandstone quality (M_2)/g	Mass removed (M_w)/g
I	270	436.50	436.40	0.10	1054.37	1042.40	11.97
II	270	438.15	438.03	0.12	1051.31	1036.54	14.77
III	270	434.60	434.45	0.15	1058.54	1045.85	12.69

Table 9. Abrasion of GWLCC machining red sandstone at different feed speed.

Effect of mechanical feed rate on grinding performance of grinding wheel

Table 9 shows the wear of diamond grinding wheels during the processing of red sandstone at different feed speeds. Based on the data presented in Table 8, it is observed that when the rotating speed remains constant at 270r/min, an increase in feed speed leads to an accelerated wear rate of the diamond grinding wheel and a continuously increase in wear mass. Conversely, the removal mass of red sandstone initially increases before subsequently decreasing. At Grade II feed speed, the material removal mass reaches its peak value. These observations show that feed speed significantly influences both the working life of the grinding wheel and machining efficiency.

Feed speed/step	Rotating speed/(r/min)	Average current /A
I	240	1.420
II	240	1.680
III	240	1.950

Table 10. Working current of GWLCC grinding red sandstone at different feed speeds.

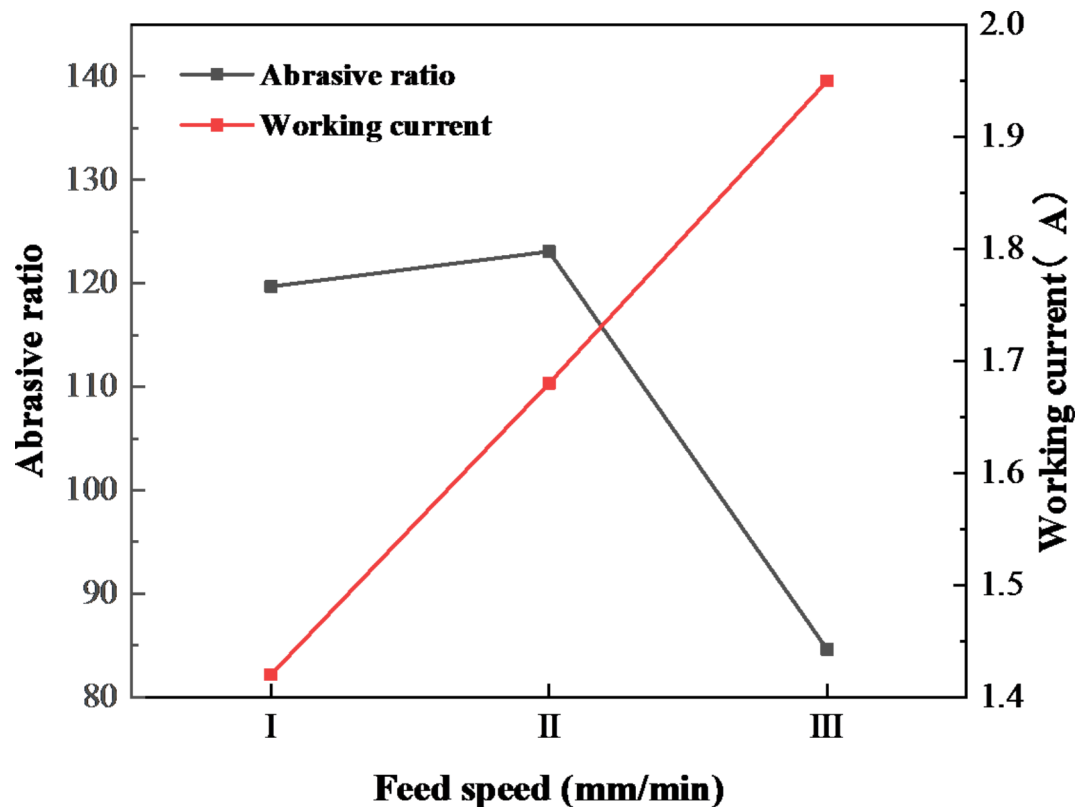


Fig. 12. Feed speed vs. abrasive ratio and current.

Table 10 shows the working current of diamond grinding wheels when grinding red sandstone at different feed speeds. The data indicates that changing the feed speed of the grinding wheel significantly affects the working current. When the speed remains constant, an increase in feed speed leads to a notable rise in working current. This observation underscores why it is essential to avoid excessively high feed speeds during the grinding work.

To more intuitively and clearly illustrate the impact of feed speed on the grinding performance of the grinding wheel, the data presented in Tables 9 and 10 have been plotted into a dotted line diagram, as shown in Fig. 12. From Fig. 12, it is evident that the abrasive ratio of the grinding wheel initially increases and then decreases with an increase in feed speed. When the abrasive ratio reaches its maximum value at Grade II feed speed, corresponding to peak grinding efficiency during this phase. However, when the feed speed is further elevated to Grade III, there is a significant decline in abrasive ratio, creating a notable disparity compared to Grade I feed speed. In addition, it is observed that the working current increases significantly with increasing feed speed, demonstrating a fundamentally linear relationship.

The test results indicate that the grinding efficiency of the diamond grinding wheel increases first and then decreases, while the grinding force continues to increase. This phenomenon occurs because the vertical pressure acting on the diamond grinding wheel escalates in tandem with an increase in feed speed. When the feed speed (vertical pressure) reaches Grade II, a closer attachment between the contact surfaces of the grinding wheel and workpiece is observed from a machining perspective. This enhanced contact effectively mitigates adverse effects caused by vibrations in the grinding equipment. From the perspective of the grinding mechanism, an optimal match between grinding speed and feed speed enhances the number of effective abrasive particles engaged in the grinding process, thereby improving overall grinding efficiency. It can be understood that when the feed speed is significantly lower than its theoretical value, the grinding working surface of the abrasive particles will detach from the workpiece surface during grinding, leading to a decrease in grinding efficiency. In such cases, increasing the feed speed will increase the number of effective abrasives involved in grinding, thereby improving the grinding efficiency. However, when the feed speed exceeds its theoretical value, increased vertical pressure

on the spindle raises friction at the grinding interface between the grinding wheel and workpiece, creating resistance to rotation. Therefore, during this grinding test, while initial increase in vertical pressure provide a beneficial force for performance enhancement, they eventually lead to negative effects. Both forces compressing against each other at the grinding interface between the grinding wheel and red sandstone result in heightened grinding resistance.

Conclusions

In this paper, we present a novel type of diamond grinding wheel with linear cooling channels, which was prepared using dual-filament FDMS 3D printing technology. The experimental material parameters and specific process were introduced in detail. At the same time, grinding tests were carried out on sandstone, concrete, alumina ceramics and optical glass using both traditional metal-bonded diamond wheels without flow channels and the newly developed diamond wheels. Further research was conducted on optimize the grinding process parameters for GWLCC. The main results may include:

- (1) A novel type of diamond grinding wheel with an innovative cooling channel was proposed, along with a technical approach for preparing this new metal-bonded diamond grinding wheel utilizing dual-filament 3D printing technology. The process flow includes batching, internal mixing, filament drawing, dual-filament 3D printing, thermal degreasing, hot-pressing sintering, and trimming. Through a comprehensive literature review and experimental validation, the formulation of diamond grinding wheel and the process parameters for dual-filament 3D printing have been established as detailed in Tables 1 and 2. Using this formulation and specified process parameters enables the production of a GWLCC characterized by a complete structure, smooth surface and uniform distribution of diamond. This confirms the feasibility of employing dual-filament 3D printing technology to manufacture diamond grinding wheels with complex structure.
- (2) Compared to the dense diamond grinding wheel, the abrasive ratio of the GWLCC increased by 32–232.67%, while the working current decreased by 6.22–14.64%. That shows that the design of linear cooling channels improves the grinding performance of the diamond grinding wheel, resulting in higher grinding efficiency and reduced grinding force. This improvement can be attributed to a smaller contact area and effective delivery of grinding fluid and efficient chip removal facilitated by the channel design.
- (3) When the feed speed is constant, the grinding efficiency of GWLCC increases with an increase in grinding speed, achieving a maximum abrasive ratio at a rotational speed of 300 r/min. Conversely, when the rotational speed remains constant, the grinding wheel abrasive ratio increases first and then decreases as feed speed rises, while the working current continues to increase. This indicates that under continuous feeding conditions, optimal grinding efficiency occurs when the thickness of material removed aligns theoretically with the feed depth, so there exists an optimal feed speed corresponding to maximum grinding efficiency for the grinding wheel. However, it is important to note that if the thickness of material removed is less than the feed depth, this may lead to potential damage to equipment. Therefore, in order to ensure test safety, a minimum feed speed was selected for certain testing phases.

Overall, the results presented above offer valuable insights into the manufacturing of grinding wheels with complicated structures. Future research is essential to investigate the porosity within the grinding wheel and the distribution of diamond particles, aiming to enhance the performance of new grinding wheel. Additionally, when conducting grinding tests using GWLCC, further studies are necessary to understand the stress distribution on the surface of the grinding wheel induced by specific structures, as well as to elucidate the unique wear mechanisms associated with diamond particles.

Data availability

Data is provided within the manuscript.

Received: 26 July 2024; Accepted: 18 November 2024

Published online: 22 November 2024

References

1. Kishore, K., Sinha, M. K. & Chauhan, S. R. A comprehensive investigation of surface morphology during grinding of Inconel 625 using conventional grinding wheels. *J. Manuf. Process.* **97**, 87–99, <https://doi.org/10.1016/j.jmapro.2023.04.053>. (2023)
2. Tian, C. C. et al. Study on formability, mechanical property and finite element modeling of 3D-printed composite for metal-bonded diamond grinding wheel application. *J. Manuf. Process.* **54**, 38–47, <https://doi.org/10.1016/j.jmapro.2020.02.028> (2020).
3. Kishore, K., Sinha, M. K., Singh, A., Archana, Gupta, M. K. & Korkmaz, M. E. A comprehensive review on the grinding process: Advancements, applications and challenges. *Proc. Inst. Mech. Eng., Part C*. **236**, 10923–10952, (2022). <https://doi.org/10.1177/09544062221110782>
4. Oliveira, J. F. G., Silva, E. J., Guo, C. & Hashimoto, F. Industrial challenges in grinding. *CIRP Ann. Manuf. Technol.* **58**, 663–680, <https://doi.org/10.1016/j.cirp.2009.09.006> (2009).
5. Teng, S. G. et al. Experimental study on grinding silicon carbide ceramics with different structured diamond grinding wheels. *Tool. Tech.* **55**, 38–43. <https://doi.org/10.3969/j.issn.1000-7008.2021.08.005> (2021).
6. Kishore, K., Chauhan, S. R. & Sinha, M. K. Investigation on grindability improvement of inconel 625 using multi-walled carbon nanotubes nanofluids-assisted minimum quantity lubrication surface grinding. *J. Mater. Eng. Perform.* <https://doi.org/10.1007/s11665-024-09648-4> (2024).
7. Ding, N., Jiang, S. N., Duan, J. S., Liu, C. & Cui, S. F. Design of new slotted structured grinding wheel. *J. Phys. Conf. Ser.* **1635**, 012013. <https://doi.org/10.1088/1742-6596/1635/1/012013> (2020).
8. Zhang, X. H. et al. Improved grinding performance of Zirconia ceramic using an innovative biomimetic fractal-branched grinding wheel inspired by leaf vein. *Ceram. Int.* **46**, 22954–22963. <https://doi.org/10.1016/j.ceramint.2020.06.070> (2020).

9. Wen, D. D. et al. Grinding performance evaluation of SiC ceramic by bird feather-like structure diamond grinding wheel. *J. Manuf. Process.* **95**, 382–391. <https://doi.org/10.1016/j.jmapro.2023.04.024> (2023).
10. Rabbey, M. *Dry Grinding with CBN Wheels, the Effect of Structuring* (University of Stuttgart, 2011). <https://doi.org/10.18419/OPUS-4429>
11. Zahedi, A. & Azarhoushang, B. Microstructuring strategies of cBN grinding wheels. *Int. J. Adv. Manuf. Technol.* **91**, 3925–3932. <https://doi.org/10.1007/s00170-017-0044-4> (2017).
12. Guo, Z. Z., Cheng, J., Wu, J., Zhang, X. F. & Liu, B. Y. Modeling and experimental study on the grinding performance of precision diamond grinding tool with defined texture. *Int. J. Adv. Manuf. Technol.* **124**, 737–758. <https://doi.org/10.1007/s00170-021-08372-w> (2023).
13. Li, K. W. et al. Preparation and performance of grinding wheel with orderly arranged abrasive particles based on laser processing technology. *Int. J. Adv. Manuf. Technol.* **127**, 4519–4532. <https://doi.org/10.1007/s00170-023-11540-9> (2023).
14. Zhang, Q. *Research on 3D Printing Spider web Structure Diamond Grinding Wheel Based on dual Filament* (Central South University, 2024).
15. He, C., Deng, H., Yuchi, G. Z., Xie, Y. M. & Yi, J. Research progress on the design and preparation of structured superabrasive grinding wheels. *Surf. Technol.* **52**, 42–56. <https://doi.org/10.16490/j.cnki.issn.1001-3660.2023.12.003> (2023).
16. Li, X. K. et al. Digital design and performance evaluation of porous metal-bonded grinding wheels based on minimal surface and 3D printing. *Mater. Des.* **203**, 109556. <https://doi.org/10.1016/j.matdes.2021.109556> (2021).
17. Nakayama, K., Takagi, J. & Abe, T. Grinding wheel with helical grooves-an attempt to improve the grinding performance. *Ann. CIRP* **25**, 133–138 (1977).
18. Azarhoushang, B., Daneshi, A. & Lee, D. H. Evaluation of thermal damages and residual stresses in dry grinding by structured wheels. *J. Clean. Prod.* **142**, 1922–1930. <https://doi.org/10.1016/j.jclepro.2016.11.091> (2017).
19. Denkena, B. et al. Enhanced grinding performance by means of patterned grinding wheels. *Int. J. Adv. Manuf. Technol.* **77**, 1935–1941. <https://doi.org/10.1007/s00170-014-6579-8> (2015).
20. Zhang, Z. Z. et al. Investigation and modeling of microgrooves generated on diamond grinding wheel by abrasive waterjet based on box-behnken experimental design. *Int. J. Adv. Manuf. Technol.* **100**, 321–332. <https://doi.org/10.1007/s00170-018-2749-4> (2019).
21. Khangar, A., Dahotre, N. B., Jackson, M. J. & Robinson, G. M. Laser dressing of alumina grinding wheels. *J. Mater. Eng. Perform.* **15**, 178–181. <https://doi.org/10.1361/105994906X95832> (2006).
22. Li, S., Peng, F. Y. & Kang, Z. X. Process experimental of laser macro-structured diamond grinding wheels for alumina grinding. *Aero Mater. Tech.* **50**, 69–74. <https://doi.org/10.12044/j.issn.1007-2330.2020.02.013> (2020).
23. Fu, Y. C. & Xu, H. J. Analysis on the temperature field on workpiece surface layer in creep feed deep grinding with slotted grinding wheel. *China Mech. Eng.* **13**, 541–543. <https://doi.org/10.3321/j.issn:1004-132X.2002.07.001> (2002).
24. Hou, Z. B., Yao, Z. Q., Sun, Y. F. & Shen, H. Grooving profile control for structured grinding wheels with picosecond pulsed laser. *Int. J. Adv. Manuf. Technol.* **119**, 5851–5862. <https://doi.org/10.1007/s00170-022-08655-w> (2022).
25. Deng, H., Xu, Z., Wang, L. Q. & Zhu, P. C. Laser micro-structuring of a coarse-grained diamond grinding wheel. *Int. J. Adv. Manuf. Technol.* **101**, 2947–2954. <https://doi.org/10.1007/s00170-018-3101-8> (2019).
26. Zhang, S. H., Tang, J., Zhou, H. & Wang, J. L. Three-dimensional printing in the application of diamond tools manufacturing. *Diam. Abras. Eng.* **38**, 51–56. <https://doi.org/10.13394/j.cnki.jgszz.2018.2.0011> (2018).
27. Wu, J. J., Zhang, S. H., Qu, F. L., Zhou, H. & Tang, J. Matrix material for a new 3D-printed diamond-impregnated bit with grid-shaped matrix. *Int. J. Refract. Met. Hard Mater.* **82**, 199–207. <https://doi.org/10.1016/j.jrmhm.2019.04.017> (2019).
28. Wu, J. J. et al. Rock breaking characteristics of a 3D printing grid-matrix impregnated diamond bit. *Int. J. Refract. Met. Hard Mater.* **89**, 1–11. <https://doi.org/10.1016/j.jrmhm.2020.105212> (2020).
29. Zhang, S. H., Su, Z., Liu, L. L., Kong, X. W. & He, T. Comparative study on ultra-thin diamond saw blades made by SLS and FDMS. *Diam. Abrasives Eng.* **41**, 38–43. <https://doi.org/10.13394/j.cnki.jgszz.2021.1.0007> (2021).
30. Su, Z., Zhang, S. H., Liu, L. L. & Wu, J. J. Microstructure and performance characterization of co-based diamond composites fabricated via fused deposition molding and sintering. *J. Alloys Compd.* **871**, 159569. <https://doi.org/10.1016/j.jallcom.2021.159569> (2021).
31. Kong, X. W. et al. Development and properties evaluation of diamond-containing metal composites for fused filament fabrication of diamond tool. *Diam. Relat. Mater.* **130**, 109423. <https://doi.org/10.1016/j.diamond.2022.109423> (2022).
32. He, T. et al. Influence of diamond parameters on microstructure and properties of copper-based diamond composites manufactured by fused deposition modeling and sintering (FDMS). *J. Alloys Compd.* **931**, 167492. <https://doi.org/10.1016/j.jallcom.2022.167492> (2023).
33. Kishore, K., Chauhan, S. R. & Sinha, M. K. Application of machine learning techniques in environmentally benign surface grinding of Inconel 625. *Tribol. Int.* **188**, 108812. <https://doi.org/10.1016/j.triboint.2023.108812> (2023).
34. Chen, J. Y., Huang, H. & Xu, X. P. An experimental study on the grinding of alumina with a monolayer brazed diamond wheel. *Int. J. Adv. Manuf. Technol.* **41**, 16–23. <https://doi.org/10.1007/s00170-008-1459-8> (2009).
35. Liu, W., Deng, Z. H., Shang, Y. Y. & Wan, L. L. Effects of grinding parameters on surface quality in silicon nitride grinding. *Ceram. Int.* **43**, 1571–1577. <https://doi.org/10.1016/j.ceramint.2016.10.135> (2017).
36. Agarwal, S. & Rao, P. V. Experimental investigation of surface/subsurface damage formation and material removal mechanisms in SiC grinding. *Int. J. Mach. Tools Manuf.* **48**, 698–710. <https://doi.org/10.1016/j.ijmachtools.2007.10.013> (2008).
37. Qiao, G. C., Zhou, M. & Wang, M. Study on surface roughness in ultrasonic vibration mill-grinding hot-pressed silicon nitride. *Adv. Mater. Res.* **500**, 269–274. <https://doi.org/10.4028/www.scientific.net/amr.500.269> (2012).
38. Deng, H., Deng, Z. & Li, S. The grinding performance of a laser-dressed bronze-bonded diamond grinding wheel. *Int. J. Adv. Manuf. Technol.* **88**, 1789–1798. <https://doi.org/10.1007/s00170-016-8902-z> (2017).
39. Luo, B., Yan, Q. S., Pan, J. S., Lu, J. B. & Huang, Z. L. Influences of processing parameters on metal-bonded diamond wheel wear when grinding a sapphire wafer. *Diam. Relat. Mater.* **113**, 108275. <https://doi.org/10.1016/j.diamond.2021.108275> (2021).

Acknowledgements

This work was supported by the Open Project of Technology Innovation Center for Deep Gold Resources Exploration and Mining, Ministry of Natural Resources (No. LDKF-2023BZX-26), the National Key Research and Development Program of China (No.2021YFB3701804), the National Natural Science Foundation of China (No.42372358, 42402325), the Natural Science Foundation of Hunan Province, China (No.2023JJ30212, 2024JJ7161), and the Research Foundation of Education Bureau of Hunan Province, China (No. 23B0579).

Author contributions

J.J. W., X. G.Z., Q. Z. and S.H.Z. wrote the main manuscript text. X.W.K. and L.L. R prepared Figs. 1, 2, 3, 4, 5 and 6; Tables 1, 2, 3, 4 and 5. Y.L.L., W. R.G and Z.J. X. prepared Figs. 7, 8, 9, 10, 11 and 12; Tables 6, 7, 8 and 9. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Q.Z. or S.Z.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2024